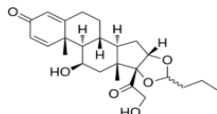


BUDEMATE NS 32
BUDESONIDE AQUEOUS NASAL SPRAY
BP 32mcg/spray × 120sprays

NAME AND STRUCTURE

NAME Budesonide Aqueous Nasal Spray BP 32mcg/spray × 120sprays

CHEMICAL STRUCTURE



MOLECULAR FORMULA $C_{25}H_{34}O_6$

MOLECULAR WEIGHT 430.5

CHARACTERISTICS

This product is an aqueous suspension of Budesonide in a suitable container fitted with an appropriate nasal delivery system.

PHARMACOLOGICAL ACTION

Budesonide is a glucocorticosteroid with a strong topical anti-inflammatory effect on the nasal mucosa and weak systemic effects after topical administration. The exact mechanism of action of glucocorticosteroids in the treatment of rhinitis is not fully understood. The activity of budesonide measured as the affinity to the glucocorticoid receptor is approximately 15 times higher than that of prednisolone. Budesonide given prophylactically prior to nasal provocation has proved to protect against immigration of eosinophils and hyper-reactivity. At recommended doses, BUDEMATE NS does not cause any clinically significant changes either in basal plasma cortisol levels or in the response to ACTH stimulation.

PHARMACOKINETICS

Absorption: The systemic availability of budesonide from BUDEMATE NS is 33% of the measured dose. The kinetics are proportional to the dose in the therapeutic dose range. The peak plasma concentration in adults from budesonide 256 mcg from BUDEMATE NS is 0.64 nmol/L and is reached within 0.7 hr. The area under curve (AUC) after administration of budesonide 256 mcg from BUDEMATE NS is 2.7 nmol × hr/L in adults and 5.5 nmol × hr/L in children, indicating a higher systemic glucocorticoids exposure in children. **Distribution and Metabolism:** Budesonide has a volume of distribution of approximately 3 L/kg. Plasma protein-binding is approximately 85-90%. Budesonide undergoes an extensive (~90%) first-pass metabolism in the liver to metabolites with low glucocorticosteroid activity. The glucocorticosteroid activity of the major metabolites, 6-β-hydroxybudesonide and 16-α-hydroxyprednisolone, is <1% of that of budesonide. Budesonide does not undergo local metabolism in the nose. Budesonide is eliminated by metabolism which is catalysed principally by the enzyme CYP3A4. The metabolites are excreted in the urine unchanged or in conjugated form via kidneys. Only negligible amounts are found unchanged in the urine. Budesonide has high systemic clearance (~1.2 L/min) and the plasma half-life after IV

administration is on average ~4 hrs.

INDICATIONS

Seasonal allergic and allergic/non-allergic perennial rhinitis. Treatment of nasal polyps.

Prevention of nasal polyps after polypectomy.

ROUTE OF ADMINISTRATION

Nasal use only

DOSAGE AND ADMINISTRATION

The posology must be adjusted individually.

Rhinitis Adults and children from 6 years: The recommended initial dose is 256 micrograms daily. The dose can be administered once daily in the morning or divided into two administrations, morning and evening. This means 128 micrograms (4 x 32 micrograms) into each nostril in the morning or 64 micrograms (2 x 32 micrograms) into each nostril morning and evening. No further effect has been seen with doses higher than 256 micrograms per day. For elderly patients the posology is the same as for adults. When the desired clinical effect has been achieved, the dose should be reduced to the lowest amount that is needed to control the symptoms. A dose of 32 micrograms in each nostril in the morning may be sufficient for some patients. In some patients relief of symptoms is already obtained within 5-7 hours after the start of treatment. The full effect is only obtained after several days of treatment (in rare cases not until after 2 weeks). Treatment of seasonal rhinitis should therefore start before exposure to the allergens. In case of severe nasal congestion an initial addition of a vasoconstrictor may be required. Supplementary treatment may sometimes be necessary in order to counteract possible ocular symptoms caused by the allergy.

Symptomatic treatment and prevention of nasal polyps: The recommended dose is 256 micrograms daily. The dose may be administered once daily in the morning or divided into two administrations morning and evening. When the desired clinical effect has been achieved, the dose should be reduced to the lowest amount that is needed to control the symptoms.

SIDE EFFECTS

Respiratory, thoracic and mediastinal disorders Common: Local irritation, epistaxis, slight haemorrhagic nasal secretion. Rare: Nasal septum perforation, ulceration of mucous membranes. Immune system disorders Uncommon: Angioedema. Urticaria, dermatitis, rash, pruritus. Very rare: Anaphylactic reaction Immediate or delayed hypersensitivity reactions including urticaria, rash, dermatitis angioedema and pruritus have been reported. In very rare cases, ulceration of mucous membrane and nasal septum perforation have occurred during use of nasally applied steroids. The cause of these side effects (the steroid, the underlying disease, or other factors) is not clear.

CONTRAINDICATIONS

Budesonide Aqueous Nasal Spray is contraindicated in patients with hypersensitivity to any of its ingredients.

PRECAUTIONS

In long-term treatment with high doses, systemic effects of glucocorticosteroids eg, hypercortisolism, adrenal suppression and/or delayed growth in children can occur. Long-term effects of nasal steroids in children have not been elucidated. Treatment with cortisone-containing medicines can lead to slower growth. Regular monitoring of growth is recommended in children and adolescents receiving

long-term treatment with corticosteroids, irrespective of the administration form. If delayed growth is suspected, this must be investigated. The benefits of glucocorticosteroid treatment should be placed in relation to possible risks of growth inhibition. Caution should be observed when treating patients with nasal fungal or herpes infections. Caution must be exercised in the treatment of patients transferred from systemically acting glucocorticosteroids to Budesonide Aqueous Nasal Spray where disturbances in the pituitary-adrenocortical function are suspected. In these patients, the dose of systemic steroid should be cautiously reduced and tests of hypothalamic-pituitary-adrenocortical function should be considered. They may also require the addition of systemic steroids in connection with periods of stress eg. surgery, trauma and others. Severely impaired hepatic function affects the pharmacokinetics of orally administered budesonide, resulting in a reduced elimination rate and increased systemic availability. Consideration may need to be given to possible systemic effects. Special care may be needed in patients with lung tuberculosis. Budesonide Aqueous Nasal Spray should not come into contact with the eyes. If Budesonide Aqueous Nasal Spray comes into contact with the eyes, rinse immediately with water. Concomitant treatment with ketoconazole or other potent CYP3A4 inhibitors should be avoided. If this is not possible, the interval between administrations of the medications should be as long as possible (see Interactions).

Instructions for Use/Handling and Disposal: When using BUDEMATE NS for the first time, the nasal applicator must be loaded with the drug. Shake the bottle, and then spray into the air, until there is an even spray. The effect of this lasts approximately 24 hrs. If a longer time elapses before the next dose is taken, the nasal applicator must be loaded with the drug again. This time, it is sufficient to spray just once into the air.

Blow nose. Shake the bottle. Remove the white protective cap.

Hold the bottle. Insert the tip into the nostril and pump the number of doses prescribed. Administer the spray into the other nostril in the same way.

Replace the brown protective cap. Do not use BUDEMATE NS nasal spray more often than prescribed. Children should only use BUDEMATE NS nasal spray under the supervision of an adult to ensure that the dose is correctly administered and that the dosage is according to the doctor's prescription.

Cleaning: Clean the upper plastic parts regularly. Remove the white protective cap and lift off the white nasal applicator. Wash the plastic parts in warm water. Allow the plastic part to air-dry completely before replacing them.

INTERACTION WITH OTHER MEDICAMENTS

Inhibitors of cytochrome P450 enzymes Ketoconazole: Ketoconazole, a strong inhibitor of cytochrome P450 (CYP) isoenzyme 3A4 (CYP3A4), the main metabolic enzyme for corticosteroids, increased plasma levels of orally ingested budesonide. Cimetidine: At recommended doses, cimetidine, a non specific inhibitor of CYP enzymes, had a slight but clinically insignificant effect on the pharmacokinetics of oral budesonide.

Co-treatment with CYP3A inhibitors, including cobicicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

IN PREGNANCY AND LACTATION: There is no data shown any generally increased risk of congenital defects in infants born to mothers treated with BUDEMATE NS by now. A slightly increased occurrence of slight heart malformation compared to what might be expected has been confirmed in infants whose mothers were exposed to BUDEMATE NS in early pregnancy, but a causal connection with the exposure is considered improbable.

During pregnancy, the aim should be the lowest effective dose and the shortest possible treatment time with BUDEMATE NS.

Budesonide is excreted in breast milk. However, at therapeutic doses of BUDEMATE NS no effects on the suckling child are anticipated. BUDEMATE NS can be used during breast-feeding.

OVERDOSAGE: Acute overdosage with this dosage form is unlikely. Chronic overdosage may result in signs/symptoms of hypercorticism.

PRESENTATION: 120 sprays per bottle, contains 32mcg Budesonide BP per spray.

STORAGE: Store below 30°C. Do not freeze. Discard in 2 months of initial use.

SHELF LIFE: 24 months.

MANUFACTURER

Jewim Pharmaceutical (Shandong) Co., Ltd.

Tai'an High-Tech Industrial Development Zone, Shandong Province, PRC.

PRODUCT REGISTRATION HOLDER IN MALAYSIA

Medispec (M) Sdn Bhd

No.55&57, Lorong Sempadan 2, Ayer Itam ,11400, Penang, Malaysia.

HOW TO USE YOUR INHALER PROPERLY



Figure 1



Figure 2



Figure 3

1. Gently blow your nose to clean the nostrils, if necessary.
2. Shake the bottle (figure 1). Remove the protective cap.
3. Hold the bottle as shown in figure 2. Before using Budesonide nasal spray suspension for the first time you must prime the nozzle (i.e. fill it with medicine). Pump the nozzle up and down several times (5-10 times), spraying into the air until an even mist is seen. The priming effect remains for approximately 24 hours. If a longer period of time passes before the next spray is taken, the nozzle must be primed (filled with medicine) again. If Budesonide nasal spray suspension is used at shorter intervals it is sufficient to spray just once into the air.
4. Insert the tip of the nozzle into your nostril as shown in figure 3 and spray once (or more if your doctor has told you to). Use the spray into the other nostril in the same way. Note, it is not necessary to breathe in at the same time as you spray.
5. Wipe the nozzle with a clean tissue and replace the protective cap.
6. Store the bottle in an upright position.

DATE OF REVISION

22 February 2018